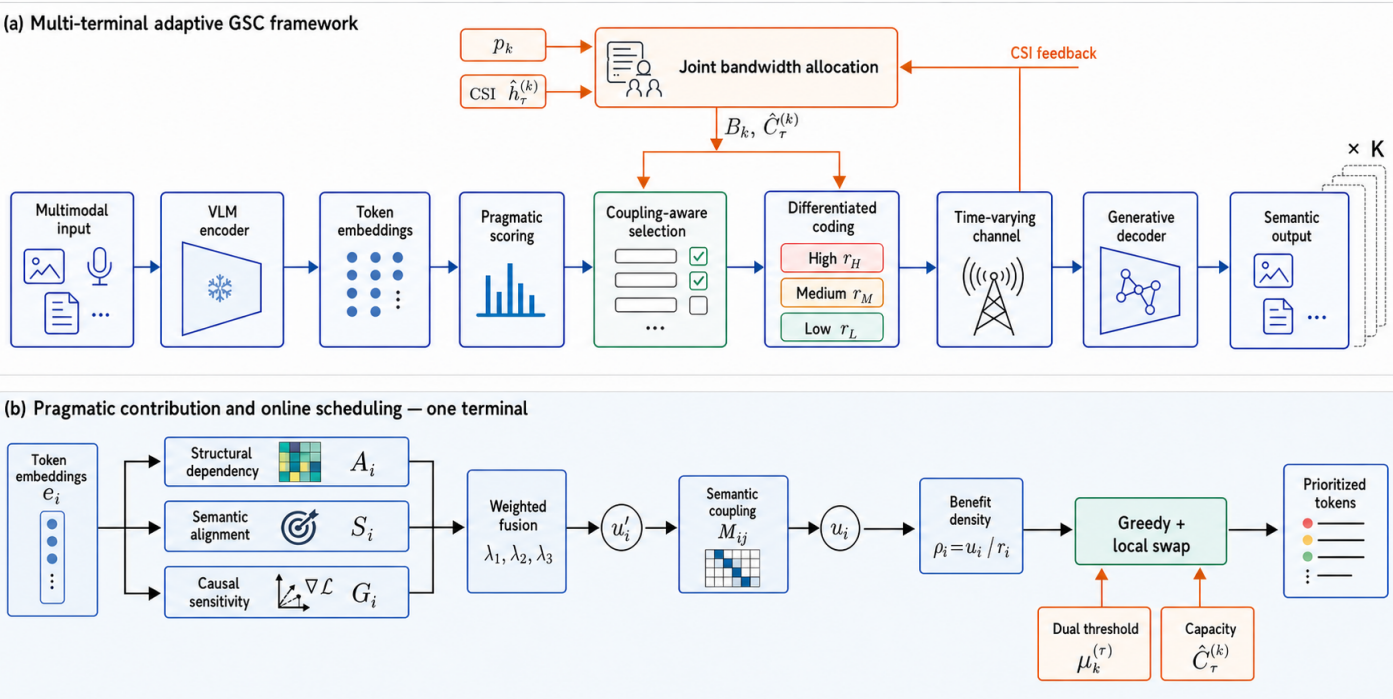

(a) Multi-terminal adaptive GSC framework
p_k
CSI $\hat{h}_\tau^{(k)}$
Joint bandwidth allocation
CSI feedback
$B_k,\ \hat{C}_\tau^{(k)}$
× K
Multimodal input
VLM encoder
Token embeddings
Pragmatic scoring
Coupling-aware selection
Differentiated coding
High r_H
Medium r_M
Low r_L
Time-varying channel
Generative decoder
Semantic output
(b) Pragmatic contribution and online scheduling — one terminal
Token embeddings e_i
Structural dependency A_i
Semantic alignment S_i
Causal sensitivity $\nabla\mathcal{L}$ G_i
Weighted fusion $\lambda_1, \lambda_2, \lambda_3$
u_i'
Semantic coupling M_{ij}
u_i
Benefit density $\rho_i = u_i / r_i$
Greedy + local swap
Prioritized tokens
Dual threshold $\mu_k^{(\tau)}$
Capacity $\hat{C}_\tau^{(k)}$